

ME 212: Product Manufacturing Technology

Course Code	ME 212
Course Name	Product Manufacturing Technology
L-T-P-C	2-0-3-3
Prerequisites	None
Intended for	UG (Core for B.Tech Mechanical)
Approval	53rd BoA (Dec 2023)
Recorded from	IC 141 by the 64th BoA (May 2026)

Preamble

Product Manufacturing Technology (PMT) combines market requirements, technological capabilities, and resources to define new products, their designs and the requisite manufacturing and field support processes. This course will cover product manufacturing technology related to metals, ceramics and polymers. Experiments will provide hands on experience w.r.t. various product manufacturing technologies.

Course Modules

- **Introduction:** Engineering materials, their classification, manufacturability and applications (5 Hours)
- **PMT for Structural Applications (Metals & Alloys):** Casting (sand casting, permanent mold casting, investment casting), forming (Rolling, Forging, Extrusion, Sheet metal operations), machining (drilling, lathe, milling), joining of metals (welding, riveting, nut-bolt assembly etc) (8 Hours)
- **PMT for Light weight applications (Polymers):** Classifications of plastics, blow molding, injection molding, extrusion, compression molding, vacuum forming, Additive manufacturing (3D printing and its types), laser machining, joining methods for plastics. (6 Hours)
- **PMT for High Temperature Applications (Glasses and Ceramics):** Powder manufacturing, mixing and blending, compacting, Sintering (with SPS), hot iso-static pressing, glass blowing (4 Hours)
- **PMT for High Performance Applications (Composites):** Compression molding of composites (3 Hours)

Laboratory Modules

S.No.	Equipment	Experiment	Turns
1.	Lathe	Turning and grooving operations on mild steel rod	1
2.	Milling, shaper machine	Facing operations on mild steel block	1
3.	Compression molding	To demonstrate manufacturing of composites	1
4.	Fitting	To make fillet, chamfer, drilling and tapping on mild steel flat sample	1
5.	Welding	To perform arc welding, gas welding and spot welding and FSW	1
6.	Sheet metal	To perform shearing, bending and riveting of galvanised iron sheet	1
7.	Foundry	To sand cast an aluminium rod	1
8.	CNC lathe and milling	To machine objects using CNC machining processes	1
9.	Laser machining	To demonstrate laser machining of an acrylic sheet	1
10.	Glass blowing	To demonstrate blowing of glass	1
11.	Additive manufacturing	To create objects using 3D printing processes	1
12.	Injection molding	To understand the plastic injection molding process	1
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Text Books

1. Groover, M.P., 2020. *Fundamentals of modern manufacturing: materials, processes, and systems*. John Wiley & Sons.
2. Kalpakjian, S. and Schmid, S.R., 2018. *Manufacturing engineering and technology*, 2001. New Jersey: Prentice Hall.

References

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